# Module 3

# Introduction to Computational Methods for the Brain Sciences

## Module Overview

**Cynthia Sandor**

Edmond J Safra Assistant Professor
UKRI Future Leader Fellow
UK Dementia Research Institute Group Leader

UK Dementia Research Institute
Imperial College London
THE MICHAEL J. FOX FOUNDATION FOR PARKINSON'S RESEARCH
Founding funders:
UKRI Medical Research Council
EDMOND J. SAFRA PHILANTHROPIC FOUNDATION

# Welcome to Module 3 — Introduction to Computational Methods for the Brain Sciences

## Course Overview

- Hands-on introduction to Python/R for brain and health data
- Foundations of **data handling, visualisation, and reproducible workflows**

## Objectives

- **Understand computational techniques** used in brain sciences
  - Demystify tools and avoid “black box” thinking
  - Focus on *doing*, not just listening
- **Build core programming skills**
  - Programming fundamentals
  - Data manipulation and analysis
  - Basic pipeline development
- **Develop scientific reasoning**
  - Hypothesis formulation and testing
  - Basic inference and interpretation
- **Learn in context: Parkinson’s disease**
  - Work with clinical and omics-related datasets relevant to brain sciences

# Workshops Overview

- 4 workshops led by **Cynthia** Sandor (Python, time-series, stats, ML)
- 2 workshops led by **Nathan Skene** (R + single-cell)
- 1 workshop led by **Martina Di Simplicio** (psychiatry) / Guest Session **Nir Grossman / Dr Nurun Fancy/ Dr Anna Mallach/ Dr Eugene Duff**

Final **Hackathon** for assessment and integration of all skills

## Lecturers



**Dr Cynthia Sandor**



**Dr Nathan Skene**



**Dr Martina Di Simplicio**



**Dr Nir Grossman**



**Dr Nurun Fancy**



**Dr Anna Mallach**



**Dr Eugene Duff**

UK Dementia Research Institute

# Workshops Details

**Thursday 5 November** – Introduction to Python
**Friday 6 November** – Introduction to Parkinson's diease Time-Series Data (Wearable Devices)

**Monday 9 November** – Introduction to Statistical Analysis and EHR Data
**Tuesday 10 November** – Introduction to Machine Learning
**Wednesday 11 November** – Introduction to Single-Cell Datasets

**Monday 16 November** – Identifying Cell-Type Specific Effects in Genetic Disorders

**Tuesday 17 November**
**Morning:** Dr *Martina Di Simplicio — Psychiatry Applications: Cognitive Differences in Self-Harm*, followed by a lived experience interview
**Afternoon: Department of Brain Sciences — Cutting-Edge Computational Neuroscience Showcase**
•Dr *Anna Mallach — TBC*
•Dr *Nurun Fancy — TBC*
•Dr *Eugene Duff* —TBC
•Dr *Nir Grossman — TBC*

**Friday 20 November – Hackathon & Assessment**

# Workshops Details

| Day | Morning | Afternoon | Room |
|---|---|---|---|
| **Thu 5 Nov** | **10:00** Module overview (Cynthia Sandor)<br>**10:30** Installation session (Rishideep Chatterjee) | **13:00** Introduction to Python: tutorial and challenge (Katarzyna Zoltowska) | RCS1 311B (VTL 3B) |
| **Fri 6 Nov** | **9:30** Time-series data (Cynthia Sandor)<br>**10:30** Tutorial and challenge (Cecilia Rodriguez) | **13:30** Tutorial and challenge<br>**14:30** Independent practical | RCS1 311B (VTL 3B) |
| **Mon 9 Nov** | **9:30** Statistics and EHR data (Cynthia Sandor)<br>**10:30** Tutorial and challenge (Jared Maina) | **13:30** Tutorial and challenge<br>**14:30** Independent practical | SAFB G16 Lecture Theatre |
| **Tue 10 Nov** | **9:30** Machine learning (Cynthia Sandor)<br>**10:30** Tutorial and challenge (Marirena Bafaloukou) | **13:30** Tutorial and challenge<br>**14:30** Independent practical | CXGH 1.14 Glenister Lecture Theatre |
| **Wed 11 Nov** | **9:30** Single-cell datasets (Nathan Skene)<br>**11:00** Tutorial walkthrough | **14:00** Challenge | RCS1 212A (VTL 2A) |
| **Thu 12 – Fri 13 Nov** | *Private study* | | |
| **Mon 16 Nov** | **9:30** Cell-type specific effects (Nathan Skene)<br>**11:00** Tutorial walkthrough | **14:00** Challenge | CXRB 116A (Collaborative Learning Suite A) |
| **Tue 17 Nov** | **9:30** Psychiatry applications (Martina Di Simplicio)<br>**11:15** Lived experience interview | **14:30** Research showcase (Nir Grossman, Anna Mallach, Nurun Fancy, Eugene Duff) | CXRB 116A (Collaborative Learning Suite A) |
| **Wed 18 – Thu 19 Nov** | *Consolidation and private study* | | |
| **Fri 20 Nov** | **9:30–17:00 Hackathon and assessment** | | CXRB 116A (Collaborative Learning Suite A) |

# Module Structure led by me: W1 to W4



Dr Cynthia Sandor

**Learning Philosophy**

- Learn by doing: hands-on coding, real data, real problems
- Each day functions as a stand-alone workshop

**Daily Schedule**

- 09:30 – 10:30 | Morning lecture: workshop introduction
- 10:30 – 14:30 | Programming tutorial and challenge (lunch 12:30 – 13:30)
- 14:30 – 16:00 | Independent practical: work on your own, no solutions provided
- 16:00 – 17:00 | Private study: consolidate learning
- Thursday 5 November: overview and installation in the morning

**Format**

- Programming tutorial: learn coding tools for different data types
- Challenge: apply what you learned to real data

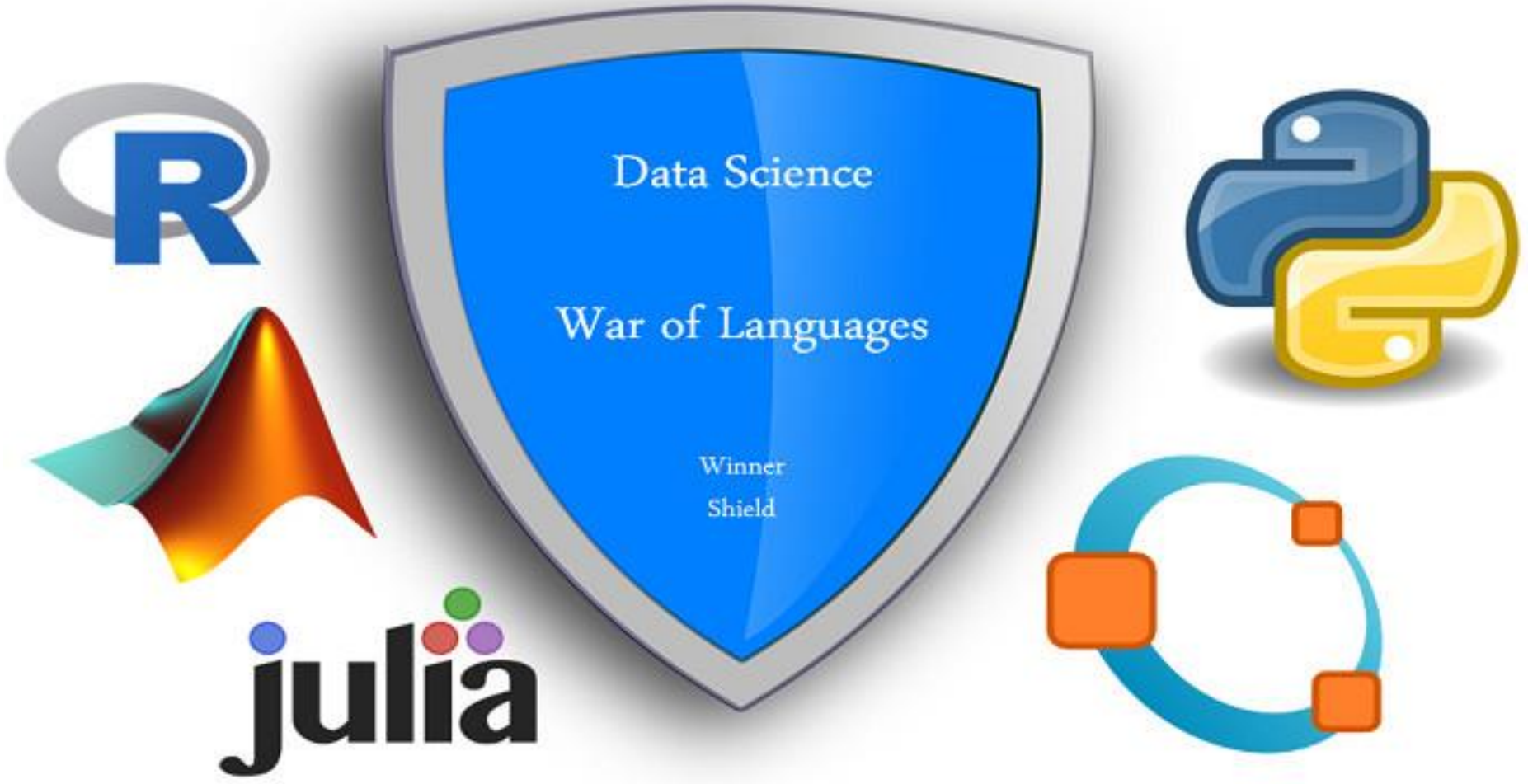
**Support**

- Work in groups of ~8 to encourage discussion and idea-sharing
- Code independently during challenge blocks
- TAs in person to answer questions — please make use of them
- Use Microsoft Teams to ask questions between sessions

**Solutions**

- Notebook solutions shared the following day on Teams (MT)

# Python vs MATLAB vs R…

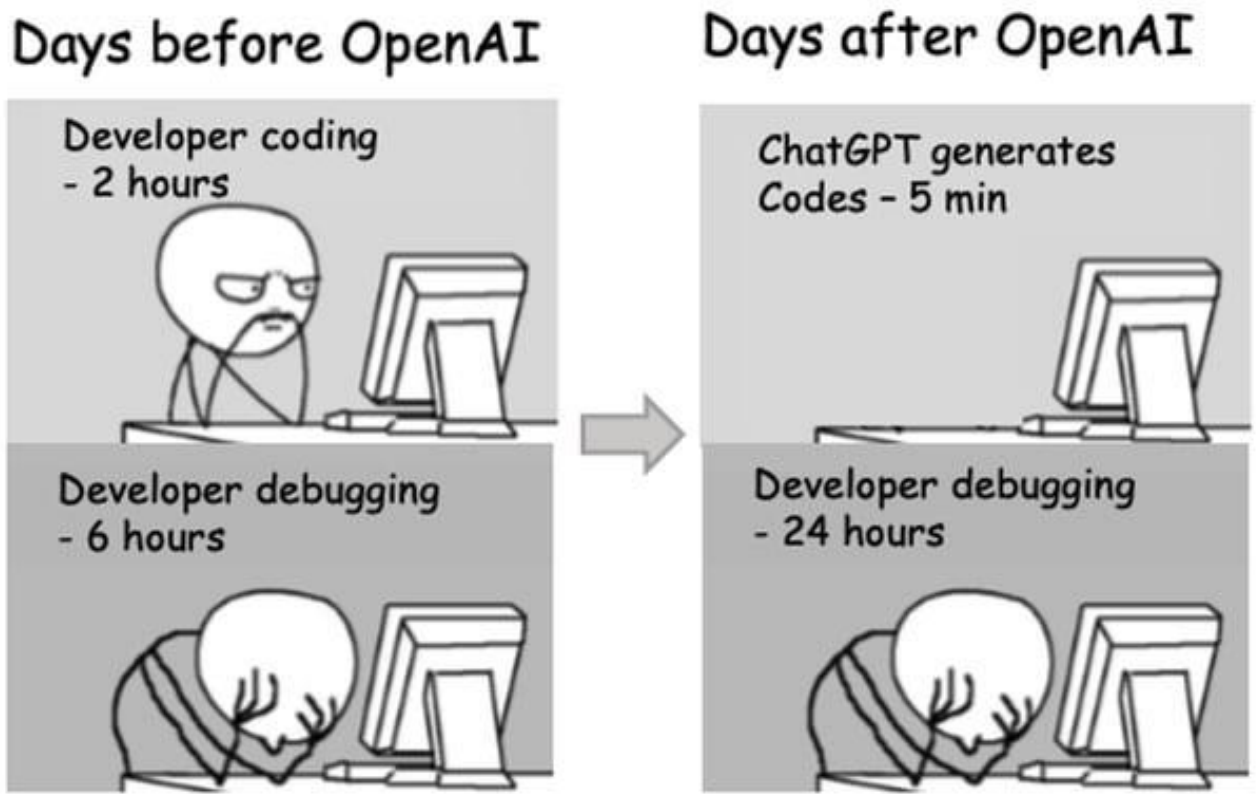


| Programming Langauge | Standouts | Setbacks | Key Libraries | Execution Speed | Learning curve | Data Analytics Capabilities | Graphical Capabilities | Tools (IDEs, Plugins, etc) | Community Support | Integration with External apps | Job market | Score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R | Open source; Good for statistical analysis and data processing; Huge collection of algorithms available as packages; Visualization support; | Steep learning curve; obscure commands | gbm, RTextTools, dplyr, zoo, ggplot2, caret | 3 | 1 | 5 | 4 | 4 | 4 | 3 | 4 | 28 |
| Python | Open source; Easy to learn; All the benefits of general-purpose programming language; Big Data ready; | Speed of execution; needs to handle library dependancies if migrated from 2.x to 3.x | Scikit-learn, Pandas, matplotlib, NumPy, SciPy, theano, nltk | 4 | 4 | 3 | 3 | 3 | 5 | 5 | 5 | 32 |
| MATLAB | Good with mathematical processes, complex matrix operations; Broad range of machine learning, signal processing and image processing libraries as toolboxes; | Proprietary; Lack of good open source ecosystem; Difficulty when the data can't be represented in matrices | Statistics and Machine Learning; Image Processing; Signal Processing; Optimizatio n; Wavelet; | 2 | 3 | 4 | 4 | 5 | 3 | 2 | 2 | 25 |
| Octave | Open source; Good for numerical computations; Built with MATLAB compatibility; Known as Clone of Matlab; Good for building preliminary models; | Lack of interoperability with external data sources - csv, databases, etc | libsvm, shogun, liblinear, ltfat, vlfeat | 2 | 2 | 3 | 2 | 2 | 3 | 2 | 1 | 17 |
| Julia | Open source; Desinged to handle numerial and scientific computing; Good performance;Ability to call C, Python functions; | Relatively new programming language; doesn't have much to offer in the way of extensive libraries | MLBase, MLUtils, MLKernels, Clustering, MachineLear ning | 5 | 3 | 4 | 2 | 2 | 2 | 3 | 1 | 22 |

Siva Prasad Katru

# Why a Learning Programmer Should Not Rely Too Much on ChatGPT or Claude

1. Loss of Problem-Solving Skills
2. Shallow Understanding
3. Risk of Bad Habits
4. Reduced Confidence
5. Missing the Broader Process
6. Curiosity Erosion
7. Ethical & Professional Risks



# Code. Break it. Fix it. Repeat.




UK Dementia Research Institute

# My Team

*My research group developed the tutorials and the hackathon.*



**Dr Katarzyna Zoltowska**



**Marirena Bafaloukou**



**Cecilia Rodriguez**



**Rishideep Chatterjee**



**Jared Maina**



**Ratnesh Maheswaran**

*Teaching assistants: answer questions and assess the hackathon*



**Antigone Fogel**



**Iona Biggart**



**Owen Brook**



**Toby Clark**



**Alexandra Kulbaka**



**Julia Borella**



**Marco Reed**



**Ruomeng Fan**

UK Dementia Research Institute

# Your Group



UK Dementia Research Institute

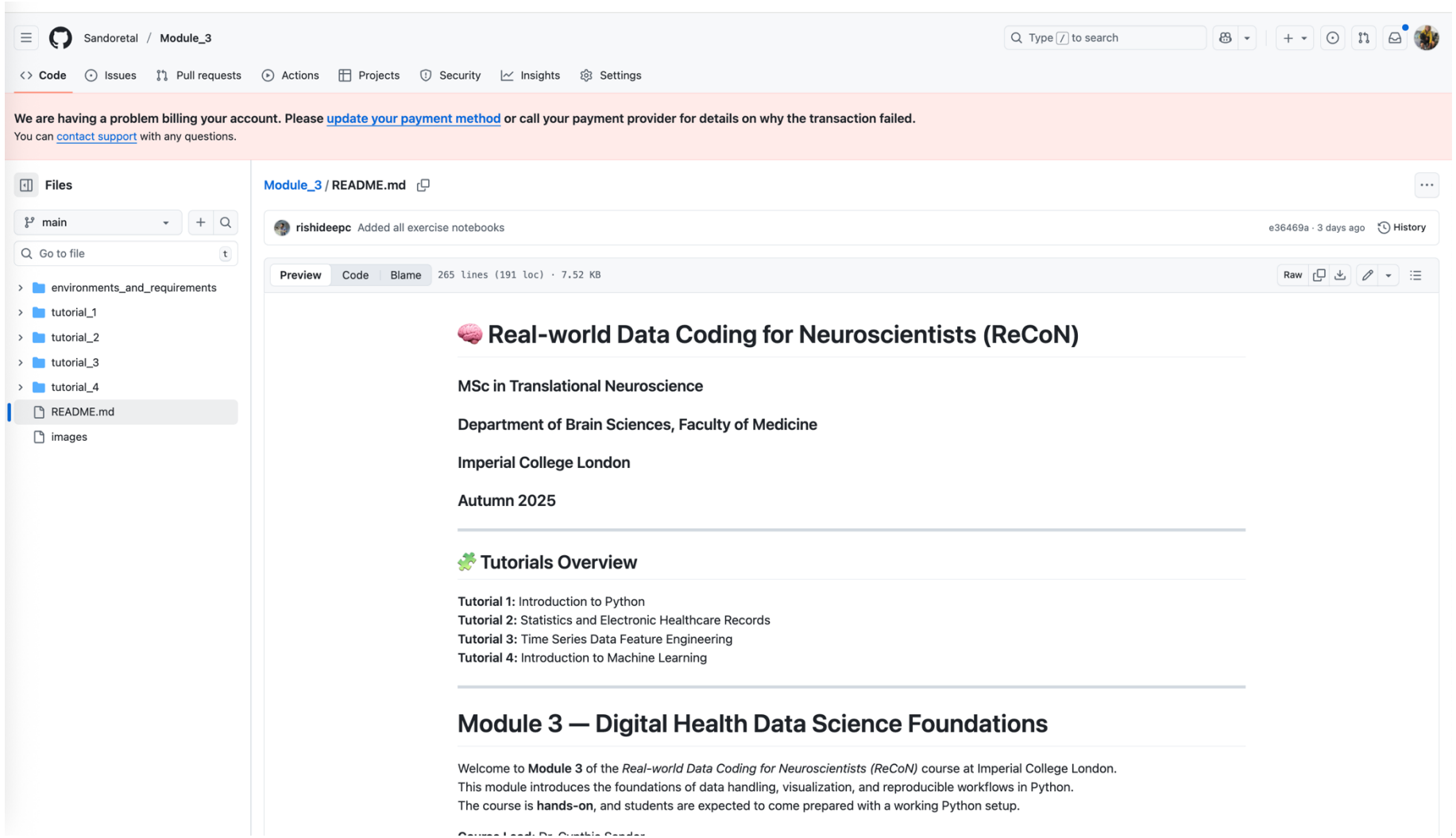
# Tutorial Groups vs Hackathon Groups

| | Tutorial groups | Hackathon groups |
|---|---|---|
| **How they are made** | Balanced by laptop operating system | Random |
| **When you find out** | Today, before the installation session | On the day: Friday 20 November |
| **What they are for** | Installing the software, discussing the tutorials and helping each other | Answering the mission's questions together and presenting the results |
| **How you work** | Discuss together; code on your own during the challenges | Work only within your group |
| **Assessed?** | No | Yes: group code 20% and group presentation 30%; the individual slides (50%) are done on your own |

**Your tutorial group is for learning; your hackathon group is for the assessment.**

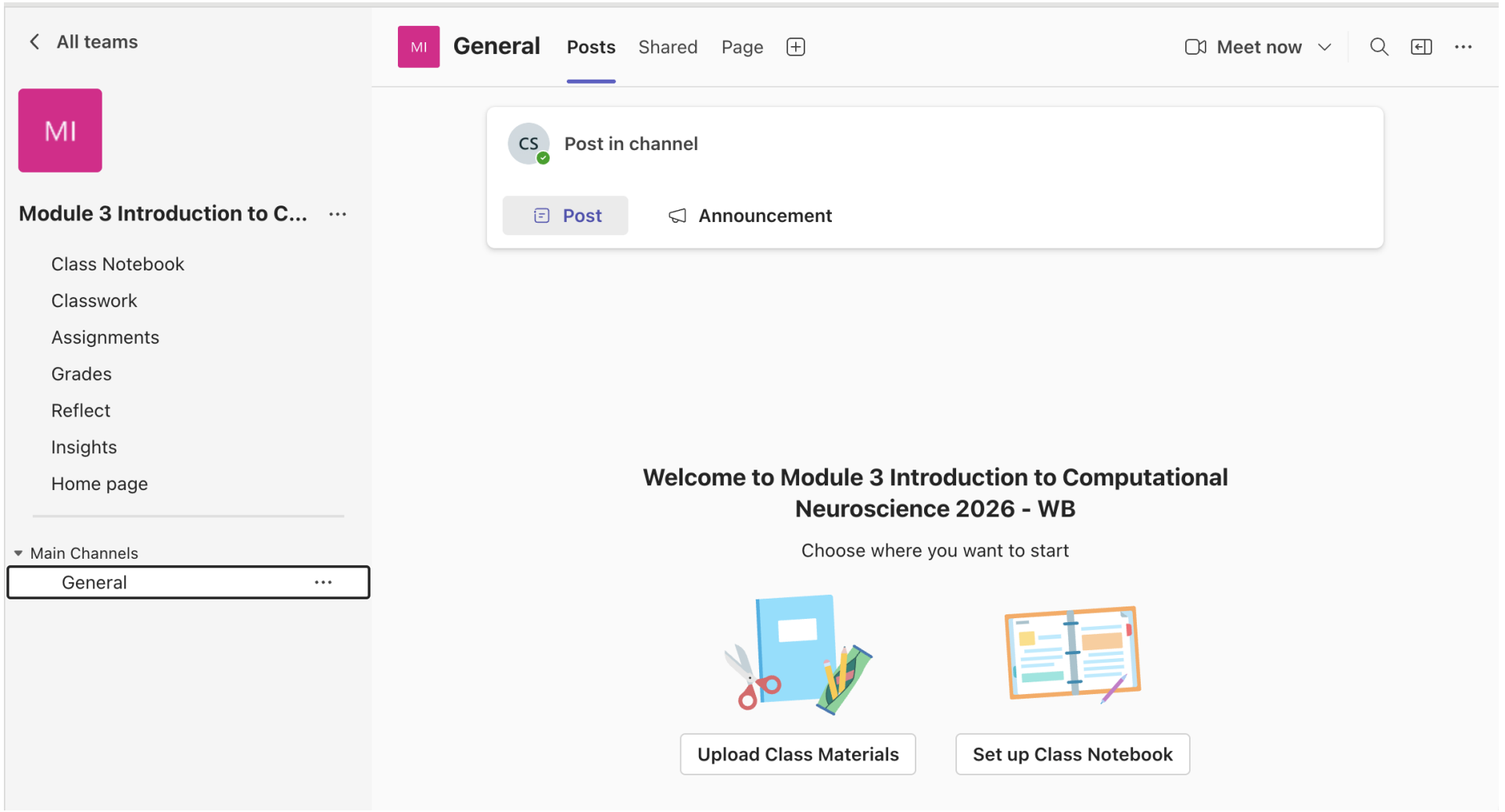
# Where to find Tutorial & Challenge?

https://github.com/Sandoretal/Module3_2026




UK Dementia Research Institute

# Microsoft Teams Class






**Dr Cynthia Sandor**

- Questions between sessions
- Solutions – Next Day

UK Dementia Research Institute

# Assessment- Hackathon

Performance in solving a computational research problem under time pressure.




Assessment includes both individual and group components.

UK Dementia Research Institute

# Hackathon- Mission Example

*Longitudinal questionnaire data have been collected from >20,000 people at two timepoints. Half the participants were surveyed early in 2020 and then in Christmas 2020–21. The other half were surveyed in May 2020, mid lockdown, and then again in Christmas 2021, mid resurgence. A subset of the measures taken have been curated for you to analyse.*

**Work in groups to address scientific questions from the data**

- Select a combination of questions to address
- Marks will relate to the number and difficulty of questions answered
- You have full access to the Internet and your notes during this assessment **but should not work across groups**
- Recommendations include (**but are not limited to**):

**Easy** | What demographic has been sampled in this study?
**Intermediate** | How does mental health relate to age & other variables at baseline or during the pandemic?
**Intermediate** | Has mental health changed during the pandemic?
**Difficult** | Which sub-populations have been most affected?
**Advanced** | To what degree can mental health be predicted from demographic and lifestyle variable



*You can grab food/drinks/coffee at any point, and we'll also take a lunch break*

# Hackathon- Schedule (Friday 20 November)

**Stage 1 — 09:30–09:50**
**Individual Planning**
**•Task: Prepare and submit 2 individual slides 50%**
•Submission: via MT to TA

**Stage 2 — 09:50–14:00**
**Group Analysis**
•Task: Collaborative coding and data analysis
We will **provide a semi-empty iPython notebook via Github**
*(includes structure, headings, and cells — but students must fill in analysis)*
**•Output: Submit group analysis code MT to TA 20%**
•Note: Coding stops at deadline

**Break 14:00–14:30**

**Stage 3 — 14:30–15:00**
**Group Slide Preparation**
•Task: Create **final presentation slides 8**

**Stage 4 — 15:00–16:00**
**Group Presentations**
**•Format: 8 min presentation + 2 min Q&A 30%**

# Today

| Day | Morning | Afternoon | Room |
|---|---|---|---|
| **Thu 5 Nov** | **10:00** Module overview (Cynthia Sandor)<br>**10:30** Installation session (Rishideep Chatterjee) | **13:00** Introduction to Python: tutorial and challenge (Katarzyna Zoltowska) | RCS1 311B (VTL 3B) |